
DEEP LEARNING LAB COURSE PROJECT REPORT

Customer Churn Analytics & Retention Suite
Multi-Layer Perceptron (MLP) Neural Network Portal

SUBMITTED BY:

Name: Tanishq Jain

Roll No: A-48

PRN: 2324000422

Class: TY CSE (AIML)

Subject: Deep Learning Lab (DLL)

Date: May 30, 2026

1. Abstract & Data Processing Pipeline

1.3 Preprocessing Pipeline (Based on Lab Assignment 3)

To feed the tabular records into the MLP neural network, the data was preprocessed via the following steps:

? TotalCharges Data Cleanup: The TotalCharges field was parsed into numerical values. Empty entries (which correspond to newly joined customers with 0 months tenure) were identified and imputed using the dataset's global mean TotalCharges value (\$2,283.56).

? Label Encoding: All 15 categorical variables (e.g. Contract, PaymentMethod, InternetService) were transformed into integer codes using LabelEncoder. This allows the inputs to be fed as numerical values while retaining distinct feature identities.

? Feature Scaling: To ensure standard gradients during backpropagation and to prevent features with large numeric magnitudes (e.g., TotalCharges) from dominating, all features were scaled using StandardScaler (mean = 0, variance = 1).

? Train-Test Splitting: The preprocessed dataset was split into an 80% training set (1,200 samples) and a 20% test set (300 samples) to ensure accurate cross-validation.

2. Neural Network Architecture & Performance

Layer Type	Nodes	Activation	Parameters
Input Layer	19	None	.
Hidden Layer 1 (Dense)	32	Sigmoid (Logistic)	640
Hidden Layer 2 (Dense)	16	Sigmoid (Logistic)	528
Output Layer (Dense)	1	Sigmoid (Logistic)	17

2.3 Experimental Performance & Convergence

During training, the binary cross-entropy loss decreased monotonically from 0.7022 (Epoch 1) to 0.4633 (Epoch 50). The evaluation of the final MLP model on the independent test set (300 samples) produced the following results:

- ? Final Test Accuracy: 81.33% (Significantly outperforming random baseline classification)
- ? Model Validation: The loss minimization curves show excellent generalization without signs of overfitting, indicating highly robust weights in the intermediate dense representations.

3. System Implementation & Conclusion
